2. CSS — Colors, Text & Box Model

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Previous: CSS Fundamentals
Next: CSS Layout

Rules stay in `styles.css`. The page is still Acme HR: one `h1`, a header, a nav, a main, one section, and a footer.

18. CSS Colors

`color` sets the text colour. `background-color` sets the surface behind it. Acme HR uses navy `#1a365d` for headings, ink `#1b2430` for text, and `#f4f6f8` for the page. Muted text must clear 4.5:1 on the page background. `#3d4654` on `#f4f6f8` clears that bar.

Example

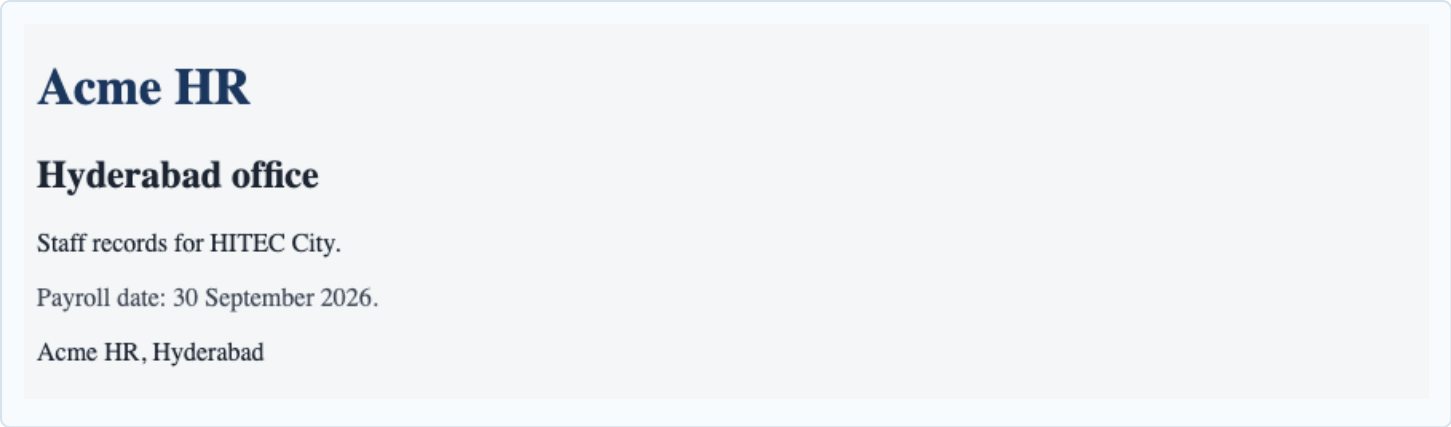
```
/* styles.css */
body {
  color: #1b2430;
  background-color: #f4f6f8;
}

h1 {
  color: #1a365d;
}

.muted {
  color: #3d4654;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p>Staff records for HITEC City.</p>
      <p class="muted">Payroll date: 30 September 2026.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



19. HEX, RGB & HSL

A colour can be written as hex, RGB, or HSL. Hex is two digits each for red, green, and blue. `rgb()` takes the same three channels from 0 to 255. `hsl()` takes hue, saturation, and lightness. These three declarations are the same navy.

Form	Acme HR navy
Hex	`#1a365d`
RGB	`rgb(26, 54, 93)`
HSL	`hsl(216, 56%, 23%)`

Example

```
/* styles.css */
h1 {
  color: #1a365d;
}

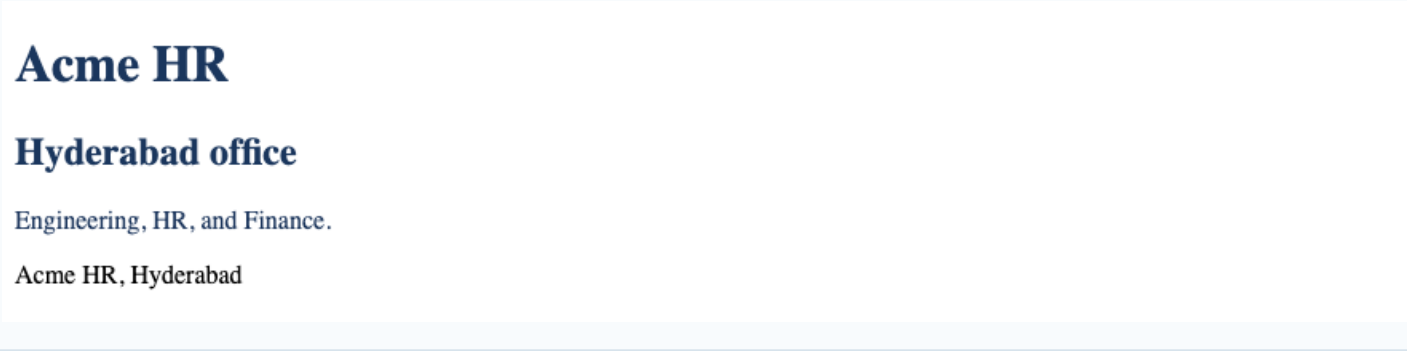
h2 {
  color: rgb(26, 54, 93);
}

.office {
  color: hsl(216, 56%, 23%);
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
```

```
      <p class="office">Engineering, HR, and Finance.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



20. Backgrounds

`background-color` fills the box. `background-image` paints a picture behind the content. `background-repeat` controls tiling. `background-position` places the image. `background-size` sets its size. A solid colour is enough for the Acme HR page. An image is an extra layer, not a replacement for the text.

Example

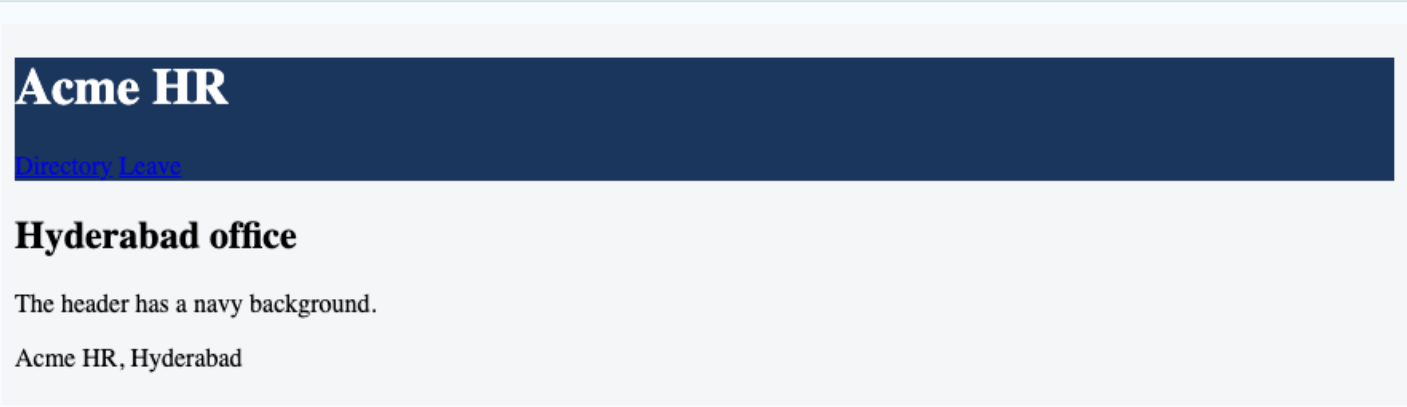
```
/* styles.css */
body {
  background-color: #f4f6f8;
}

header {
  background-color: #1a365d;
  color: #ffffff;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
    <nav>
      <a href="directory.html">Directory</a>
      <a href="leave.html">Leave</a>
    </nav>
  </header>
  <main>
    <section>
```

```
      <h2>Hyderabad office</h2>
      <p>The header has a navy background.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



21. Gradients

A gradient is a background image the browser paints. `linear-gradient` blends colours along a line. The first argument is the direction. The rest are colour stops. Use a gradient as a band behind the header. The heading text stays real text.

Example

```
/* styles.css */
header {
  background-image: linear-gradient(#1a365d, #0f6b4c);
  color: #ffffff;
  padding: 1rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p>The header blends navy into green.</p>
    </section>
  </main>
```

```
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



22. Typography

Typography is the choice of type, size, and spacing. Set a readable font on `body`. Headings and paragraphs inherit it. Color and font inherit. A line length around 60 to 75 characters stays comfortable. `max-width: 40rem` on the section keeps the office paragraph from stretching across a wide monitor.

Example

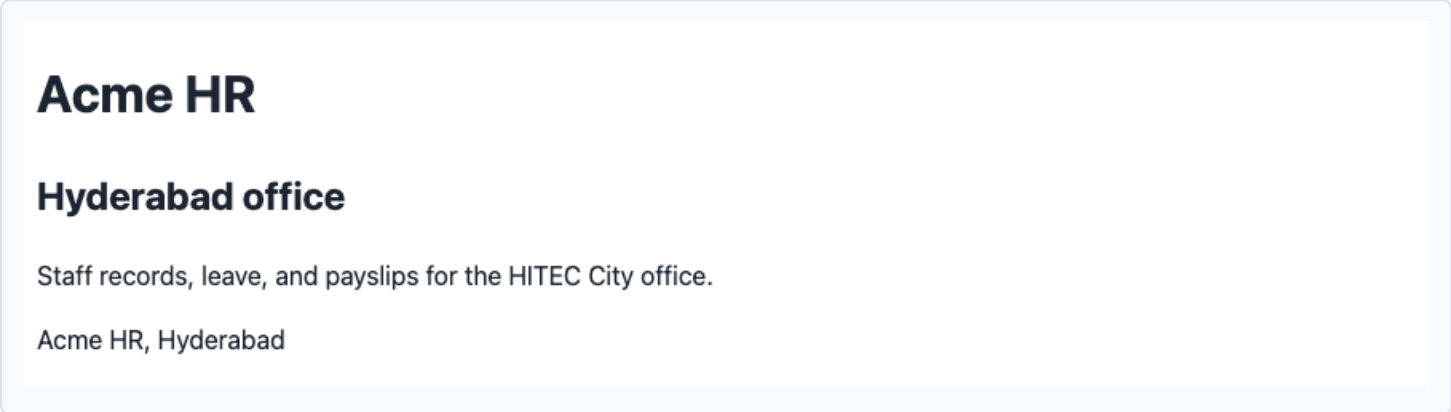
```
/* styles.css */
body {
  font-family: system-ui, sans-serif;
  line-height: 1.5;
  color: #1b2430;
}

section {
  max-width: 40rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p>Staff records, leave, and payslips for the HITEC City office.</p>
    </section>
```

```
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



23. Font Family & Web Fonts

`font-family` lists fonts in order. The browser uses the first one it has. A web font is a file the page loads. `@font-face` names that file. If the file is missing, the next name in the list is used. Django serves the font from `static/`. Spring Boot serves it from `src/main/resources/static/`.

Example

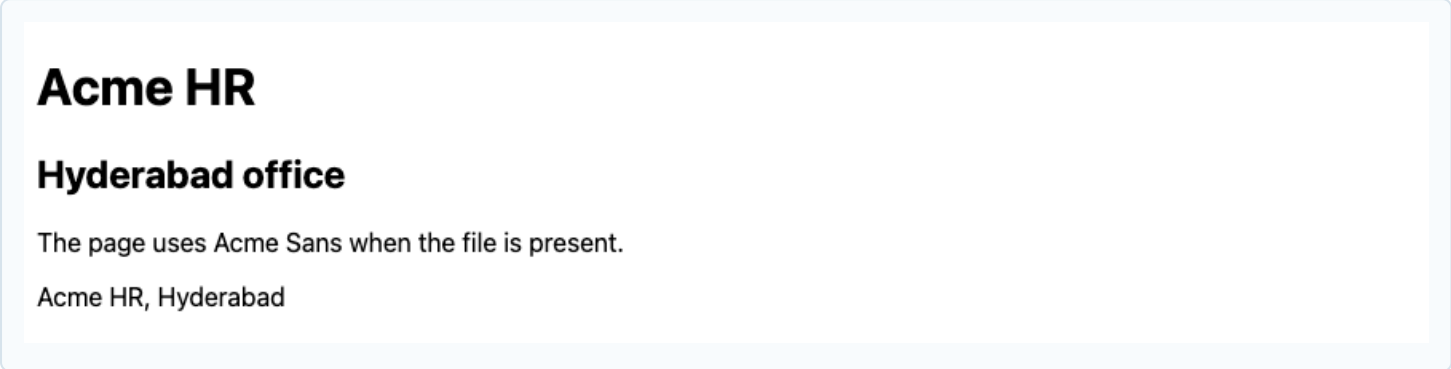
```
/* styles.css */
@font-face {
  font-family: "Acme Sans";
  src: url("fonts/acme-sans.woff2") format("woff2");
}

body {
  font-family: "Acme Sans", system-ui, sans-serif;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p>The page uses Acme Sans when the file is present.</p>
    </section>
  </main>
  <footer>
```

```
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



24. Font Size, Weight & Style

`font-size` sets the size. Use `rem` so it follows the root font. `font-weight` sets the thickness. `700` is bold. `400` is regular. `font-style: italic` slants the text. A payroll warning can be bold. A note can be italic. Importance in the HTML still belongs on `<strong>`.

Example

```
/* styles.css */
body {
  font-size: 1rem;
}

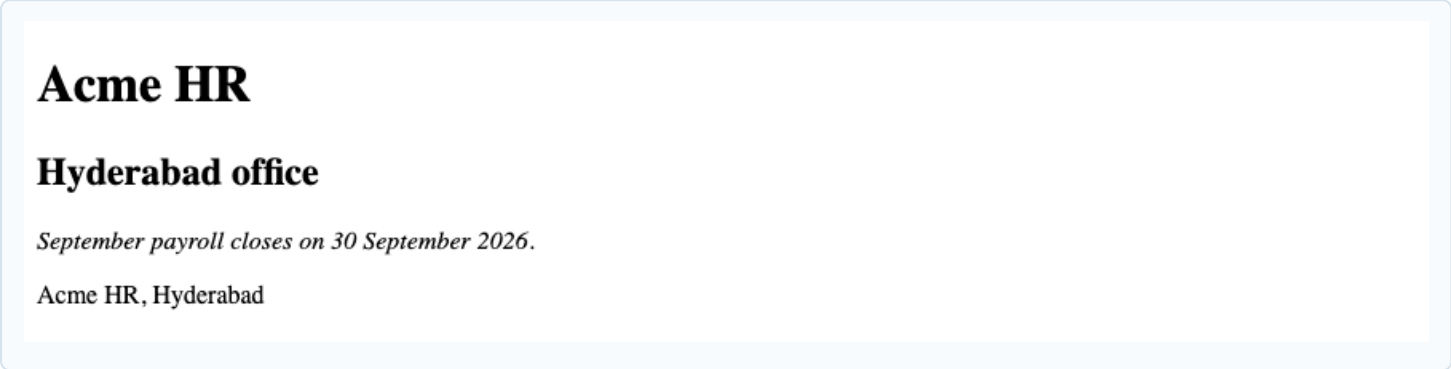
h1 {
  font-size: 2rem;
  font-weight: 700;
}

.note {
  font-style: italic;
  font-weight: 400;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p class="note">September payroll closes on 30 September 2026.</p>
    </section>
```

```
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



25. Text Alignment & Decoration

`text-align` sets the alignment of the line: `left`, `right`, `center`, or `justify`. Body text stays left aligned. `text-decoration` draws a line: `underline`, `none`, or `line-through`. Links are underlined so they are not known by colour alone. A superseded allowance can use `line-through`. The HTML `<s>` still carries the meaning.

Example

```
/* styles.css */
h1 {
  text-align: left;
}

a {
  text-decoration: underline;
}

.old-allowance {
  text-decoration: line-through;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
    <nav>
      <a href="directory.html">Directory</a>
      <a href="leave.html">Leave</a>
    </nav>
  </header>
  <main>
```

```
    <section>
      <h2>Hyderabad office</h2>
      <p>Previous special allowance <span class="old-allowance">₹12,000</span>.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



26. Line Height & Letter Spacing

`line-height` sets the space between lines. A unitless value such as `1.5` scales with the font size. `letter-spacing` adds space between characters. Body text uses `1.5`. A small heading can use a little letter spacing. Large letter spacing on a long paragraph makes it harder to read.

Example

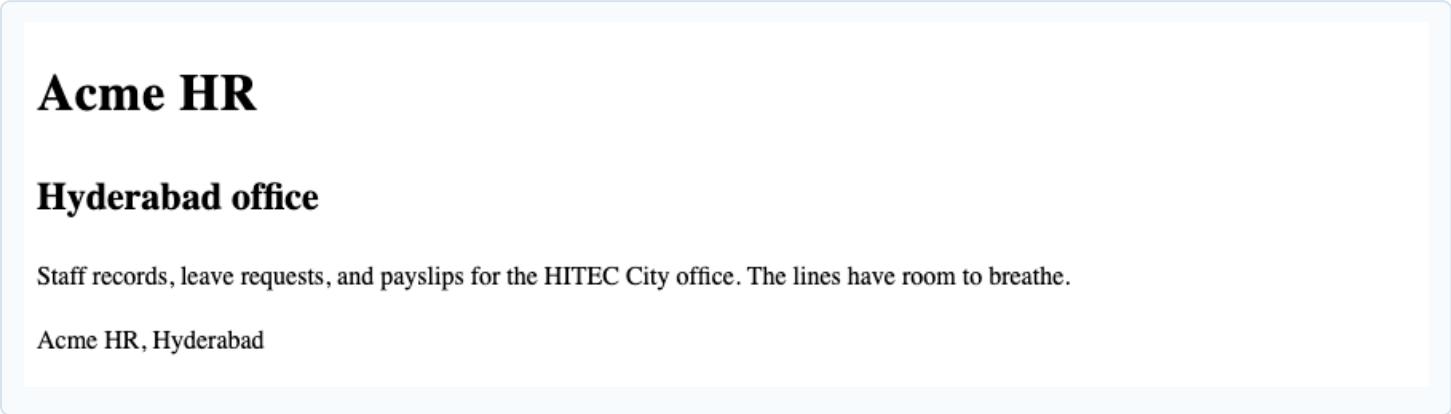
```
/* styles.css */
body {
  line-height: 1.5;
}

h1 {
  letter-spacing: 0.02em;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
```

```
      <p>Staff records, leave requests, and payslips for the HITEC City office. The lines have room to b
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



27. CSS Box Model

Every element is a box. From the inside: content, padding, border, then margin. Padding is space inside the border. Margin is space outside it. The width you set applies to the content box until `box-sizing` changes that. Margin, padding, and border do not inherit.

Layer	What it is
Content	The text or the image
Padding	Space inside the border
Border	The line around the padding
Margin	Space outside the border

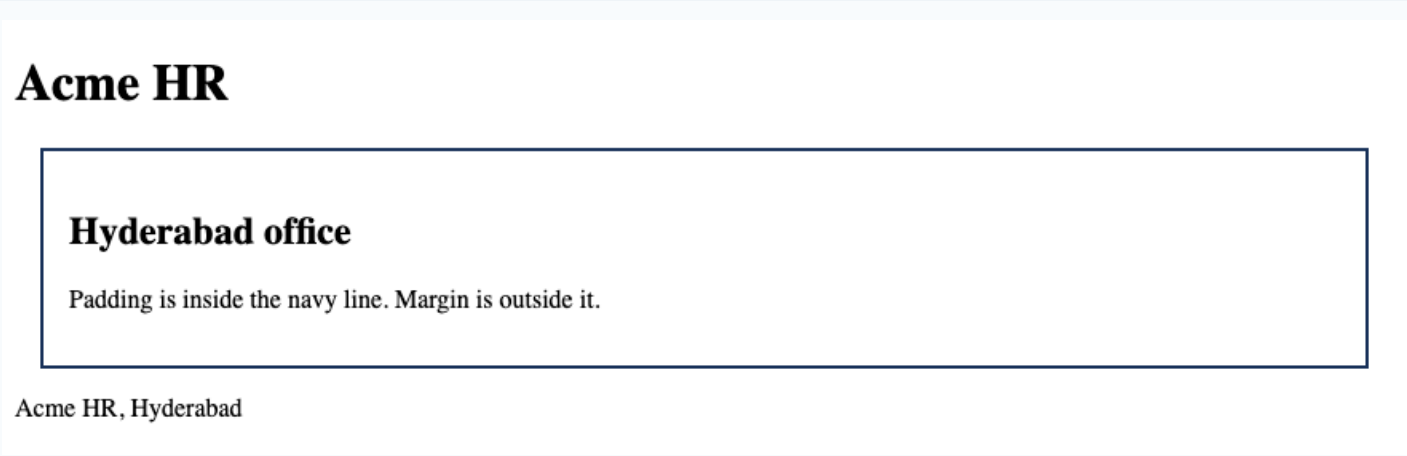
Example

```
/* styles.css */
.panel {
  margin: 1rem;
  padding: 1rem;
  border: 2px solid #1a365d;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
```

```
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section class="panel">
      <h2>Hyderabad office</h2>
      <p>Padding is inside the navy line. Margin is outside it.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



28. Width & Height

`width` and `height` set the size of the content box by default. A block stretches to the container when the width is `auto`. Height follows the content when it is `auto`. `max-width` stops a section from growing past a readable measure. Set a height only when the design needs a fixed band, such as the header.

Example

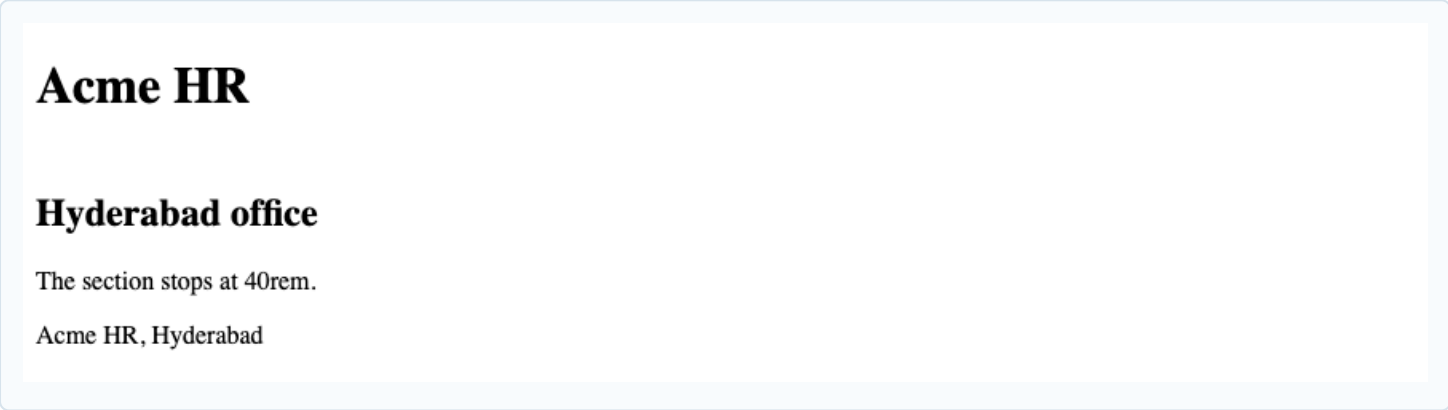
```
/* styles.css */
section {
  width: 100%;
  max-width: 40rem;
}

header {
  min-height: 4rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
```

```
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p>The section stops at 40rem.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



29. Margin & Padding

Margin separates a box from its neighbours. Padding separates the content from the border. Use `rem` for both on Acme HR. Vertical margins of two blocks can collapse: the gap becomes the larger margin, not the sum. Padding does not collapse.

Example

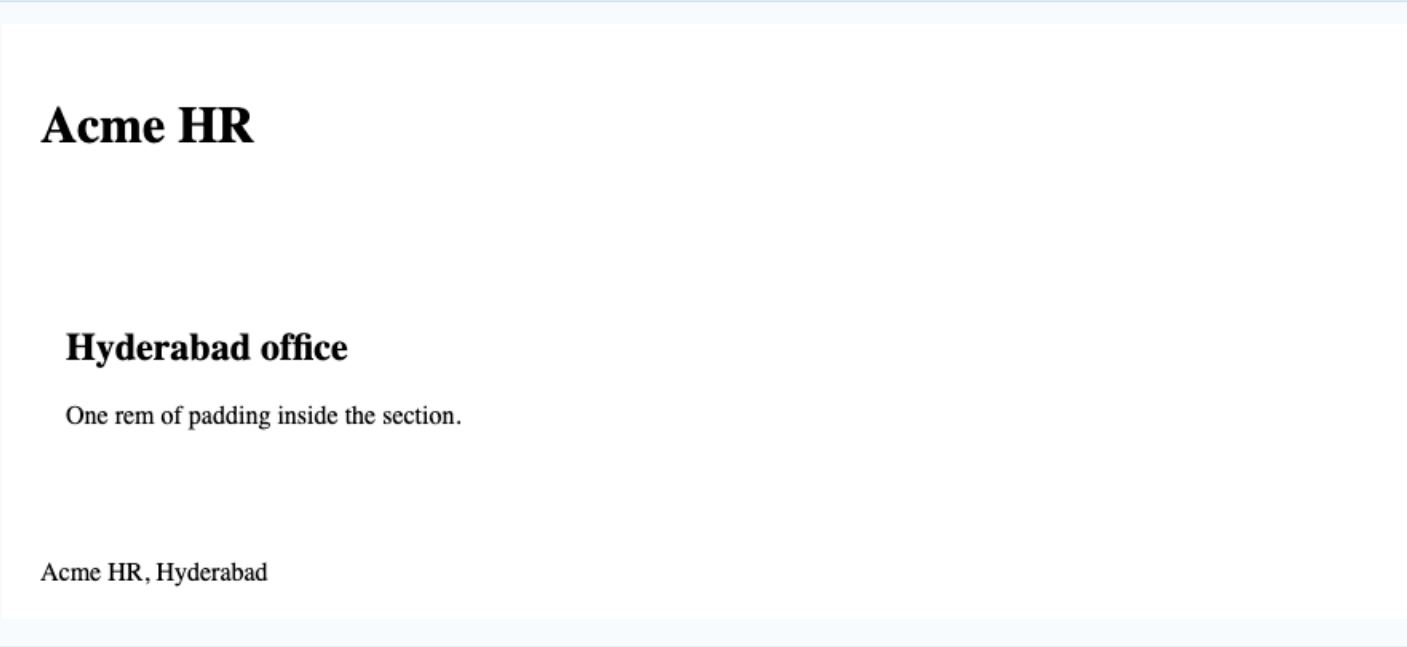
```
/* styles.css */
header, main, footer {
  padding: 1rem;
}

section {
  margin-top: 1rem;
  padding: 1rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
```

```
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p>One rem of padding inside the section.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



30. Border & Border Radius

`border` sets width, style, and colour together. `border-radius` rounds the corners. A payslip panel and a notice can use a 1 pixel solid border. A large radius makes a pill. A small radius, such as `0.25rem`, is enough for a panel.

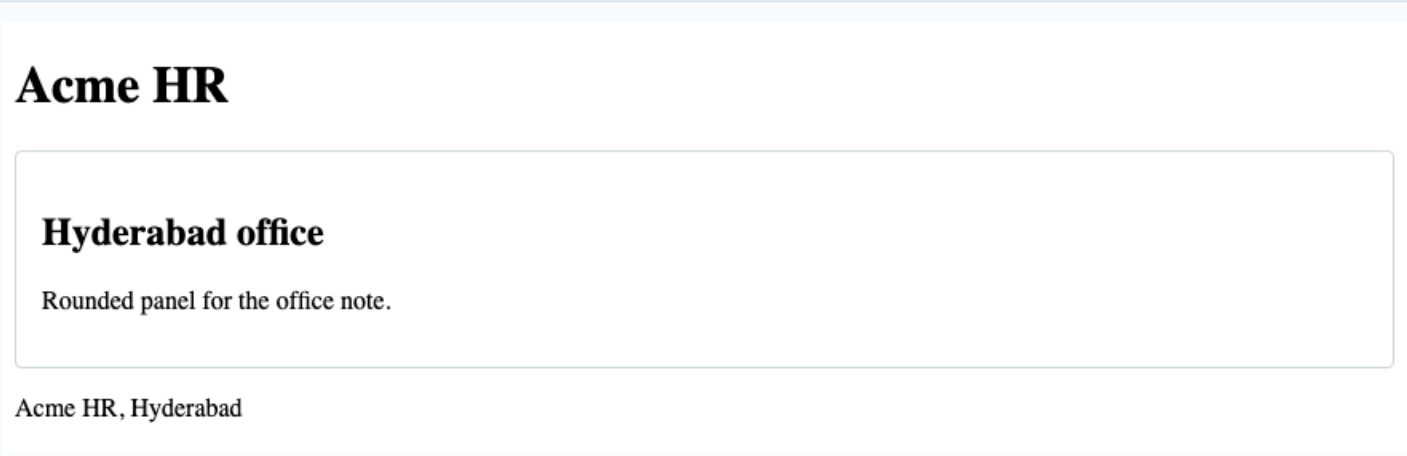
Example

```
/* styles.css */
.panel {
  border: 1px solid #d0d7de;
  border-radius: 0.25rem;
  padding: 1rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
```

```
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section class="panel">
      <h2>Hyderabad office</h2>
      <p>Rounded panel for the office note.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



31. `box-sizing`

By default, `width` is the content only. Padding and border are added on top, so the box grows past the width you set. `border-box` includes padding and border in the width. Use `*` for `box-sizing: border-box`. One rule then covers every element.

Example

```
/* styles.css */
* {
  box-sizing: border-box;
}

.panel {
  width: 20rem;
  padding: 1rem;
  border: 4px solid #1a365d;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
```

```
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section class="panel">
      <h2>Hyderabad office</h2>
      <p>The 20rem width includes the padding and the border.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



32. `box-shadow`

`box-shadow` draws a shadow outside the border. The values are horizontal offset, vertical offset, blur, and colour. A small shadow lifts a panel off the page background. The shadow is decoration. A solid border still marks the edge of the panel.

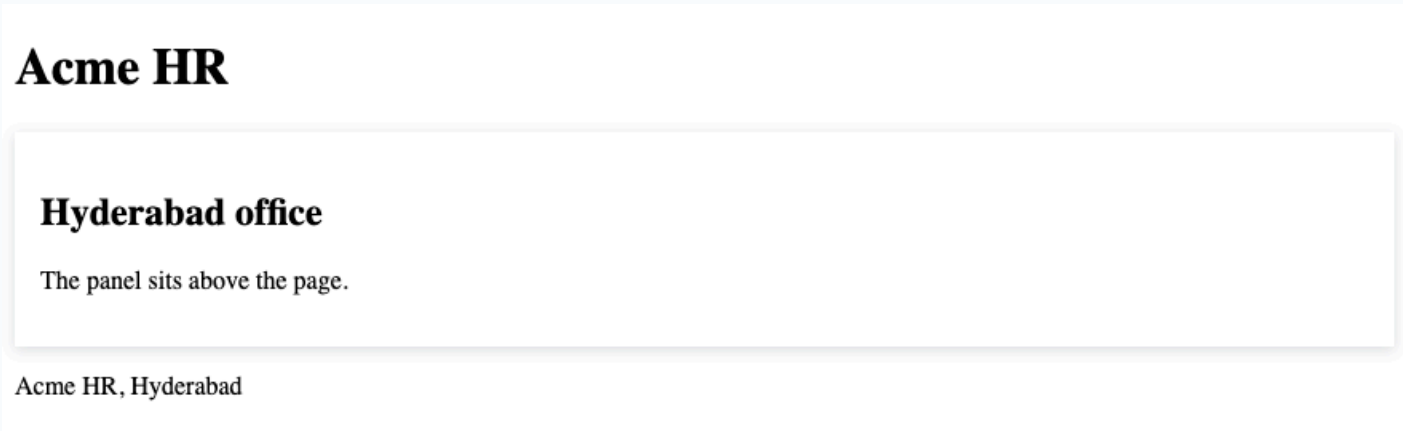
Example

```
/* styles.css */
.panel {
  background-color: #ffffff;
  padding: 1rem;
  box-shadow: 0 2px 8px rgba(27, 36, 48, 0.15);
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
```

```
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section class="panel">
      <h2>Hyderabad office</h2>
      <p>The panel sits above the page.</p>
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:



33. `overflow`

`overflow` decides what happens when content is larger than the box. `visible` lets it spill. `hidden` clips it. `auto` adds a scrollbar when it is needed. A fixed-height note on the office page uses `auto` so the rest of the paragraph can be scrolled. Do not clip text the employee must read unless the box scrolls.

Value	Result
`visible`	Content spills out
`hidden`	Content is clipped
`auto`	A scrollbar appears when needed

Example

```
/* styles.css */
.note {
  height: 3rem;
  overflow: auto;
  border: 1px solid #d0d7de;
  padding: 0.5rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <section>
      <h2>Hyderabad office</h2>
      <p class="note">September payroll includes basic pay, HRA, and special allowance. PF and professi
    </section>
  </main>
  <footer>
    <p>Acme HR, Hyderabad</p>
  </footer>
</body>
</html>
```

This will produce the following result:

